Responses to Reviewer's Comments for Manuscript

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Model-based exploration strategies can be measured convergently but are not associated with questionnaire assessments and psychiatric diagnostic criteria

Addressed Comments for Submission to

Scientific Reports

by

Kristin Witte, Mirko Thalmann, Eric Schulz

Please find enclosed the revised version of our previous submission entitled "Model-based exploration strategies can be measured convergently but are not associated with questionnaire assessments and psychiatric diagnostic criteria" with manuscript number 1682a231-a9fb-47fa-88ee-62b3bb5d6ea6. We are grateful for the comments and suggestions provided by you and the reviewers. They have substantially helped us to further improve the quality of our manuscript. In this revision, we have carefully addressed the reviewers' further comments. A summary of modifications and a detailed point-by-point response to the comments from Reviewers 1 and 2 (following the reviewers' order in the decision letter) are provided below.

Sincerely,

Kristin Witte, Mirko Thalmann, Eric Schulz

Note: To enhance the legibility of this response letter, all the editor's and reviewers' comments are typeset in blue boxes. Text from the manuscript is displayed in gray boxes. Rephrased or added sentences are typeset in bold. The tables and figures are numbered as they appear here. So the numbering might differ from the original manuscript. Furthermore, all endnotes are presented as footnotes, since this document does not have an appendix. In the main manuscript, all footnotes are numbered and presented in the appendix.

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1 Authors' Response to the Editor

1.1 General remarks

General Comments.

We are pleased to let you know that your manuscript has now passed through the review stage and is ready for revision. Many manuscripts require a round of revisions, so this is a normal but important stage of the editorial process.

Editor comments

The manuscript has now been reviewed by external referees, raising several major concerns. Please revise the mansucript, along with addressing these issues.

To ensure the Editor and Reviewers will be able to recommend that your revised manuscript is accepted, please pay careful attention to each of the comments that have been pasted underneath this email. This way we can avoid future rounds of clarifications and revisions, moving swiftly to a decision.

Response:

We would like to thank the editor and the reviewers for inviting our manuscript for resubmission. We are also grateful for all the helpful comments that helped us improve our manuscript. Below, we address every comment in detail.

We are also resubmitting a revised version of our manuscript with changes highlighted in red.

2 Reviewer 1

3 Authors' Response to Reviewer 1

3.1 General remarks

General Comments.

This paper investigates the psychometric properties of exploration strategies (i.e., directed and random/value-driven exploration) and their reliability and validity with three commonly used choice tasks: the Horizon task, the Two-armed bandit task, and the Restless bandit task. Data was collected from two online study sessions that were six weeks apart. Overall, the results suggest 1) poor to acceptable test-retest reliability, 2) a lack of convergent validity of the exploration strategies across three tasks, 3) exploration strategies across the three tasks should be considered as performance-related constructs given their high correlation with working memory (i.e., the absence of a divergent validity), and 4) poor external validity, as model-based measures of exploration strategies did not show relationships with self-reported questionnaires related to exploration or a set of affective and mood scale measures.

The focus of this paper and rigorous analytic approaches could make a useful contribution to current psychology research as well as to the field of computational psychiatry. These findings are also consistent with recent discussions in the literature that challenge the use of computational measures due to limited psychometric properties as well as poor reliability and validity. However, addressing several points would be helpful. My one major concern is that they appear to have changed the tasks and models in ways that limit comparability to previous studies, without substantial discussion or rationale. Below, I list each comment in detail, progressing from the start to end of the manuscript.

Response:

We would like to thank Reviewer 1 for the accurate summary of our manuscript and their confidence in the relevance of our contribution. Below, we will address all raised concerns in detail.

3.2 Comment 1: Definition of exploration strategies and their comparison across tasks

Point 1

In paragraph 2 of the Introduction section, the authors explained the three tasks well and briefly compared the difference in how exploration is encouraged in different task structures. However, such comparisons were not made in terms of the computational measures of exploration strategies. It would be helpful for the authors to also provide some discussion on how model-based measures of exploration are defined in the three bandit tasks. This is crucial, since the paper focuses mainly on these model-based measures. They are labelled equivalently throughout the figures, but there are some qualitative and quantitative distinctions. There is some discussion regarding these differences in the convergent validity section, but it would help to address this earlier on when providing task descriptions.

Response:

We would like to thank Reviewer 1 for drawing our attention to this critical point. We have added a paragraph to the introduction that clarifies this point. We also insert it below for your convenience.

> In addition to these differences in how the tasks motivate exploration behavior, there are also subtle differences in how exploration strategies are defined in the computational models of these tasks: In the Horizon task, the measurement of exploration is based on a difference score between the two horizon conditions. In particular, directed exploration is defined by the difference between choosing the more informative arm in the long versus the short Horizon, i.e., the degree to which participants seek out information when it can be used later in that round compared to when it cannot be. Value-guided exploration is defined by the difference between the weight relating the expected value difference between the two arms to choice probability in the long versus the short Horizon, i.e., the degree to which participants decide against the arm with the higher expected value. In the Two-armed bandit and in the Restless bandit, directed exploration and value-guided exploration are defined without calculating a difference score. That is, they are merely defined by the degree to which participants seek out uncertain options and by the weight with which they relate the expected value difference to choice probability, respectively. Typically, a different form of exploration, random exploration, is also calculated in the Two-armed bandit. Random exploration is defined by the degree to which participants' behavior gets less deterministic as overall uncertainty increases, a strategy sometimes referred to as Thompson sampling(Gershman, 2018).

3.3 Comment 2: Fewer trials in the Horizon task

Point 2

In the recoverability section, the authors stated that the Horizon task showed worse recoverability than the other two bandit tasks. Similarly, in the Reliability section, the Horizon task also showed lower reliability than the other two bandit tasks. The authors stated this could be due to the way exploration was defined (i.e., the difference between the long and short horizon conditions). However, it seems the smaller number of games administered could be another possible reason, since only the first free choice is typically fit. For the other two bandit tasks, data from all trials (N = 300 in the Two-armed bandit task and N = 200 in the Restless bandit task) are fitted to the model. In the Horizon task, in contrast, only the first free choice trials were fitted to the model (N = 80).

I agree that using the difference score between long and short horizon may be one of the reasons contributing to lower parameter recoverability, but it is worth highlighting that the number of data points included during model fitting differed also between tasks. This might be a crucial point, as previous studies that have administered the Horizon task in two study sessions for different reasons (e.g., pharmacological manipulation, sham or non-invasive stimulations) do indeed have different numbers of games than tested here. For example:

- N = 120, 1 week apart: Warren, C.M., Wilson, R.C., Van Der Wee, N.J., Giltay, E.J., Van Noorden, M.S., Cohen, J.D., Nieuwenhuis, S., 2017. The effect of atomoxetine on random and directed exploration in humans. PloS one 12, e0176034.
- N = 160, at least 5 days apart: Zajkowski, W.K., Kossut, M., Wilson, R.C., 2017. A causal role for right frontopolar cortex in directed, but not random, exploration. Elife 6, e27430.

Response:

We would like to thank Reviewer 1 for pointing out this interesting and complicated point. We had put the following considerations into selecting the number of games in the Horizon task: Since the entire study involved three bandit tasks, three working memory tasks and five questionnaires, it was already pretty lengthy, and keeping participants' attention was imperative. Since the Horizon task was the longest one of the three bandit tasks, we therefore performed a priori power analyses to test which number of games was necessary to confidently detect the expected fixed effects using different sample sizes. To this end, we fit our model to publicly available data by Zaller et al. (Zaller, Zorowitz, & Niv, 2021)(N = 99, 80 games) and simulated choices 100 times over for three different sample sizes (30, 60, 99) and four different numbers of games (40, 80, 160, 320). We found that for 80 or more games the fixed effects in the original data could be detected 100% of the time for all sample sizes. Further, several previous studies had successfully applied the Horizon task using only 80 games (Zaller et al., 2021; Smith et al., 2022). Following your comment, we extended these power analyses to also estimate correlations between original and recovered parameter estimates of the individual participants using the same dataset, a sample size of 99 participants (full sample) and 40, 80 or 160 games. We found that the correlations were high for 80

games, making us confident that 80 games should in principle be enough to obtain high reliability and recoverability estimates. You can find a table of these correlations below.

We added a sentence to the Methods section outlining these considerations. Additionally, we added a short paragraph to the discussion section, elaborating on this limitation. You can find both below.

Table 1:

Horizon	Predictor	N games	cor	lower 95%CI	upper 95%CI
short	directed	40	0.75	0.73	0.77
short	directed	80	0.81	0.8	0.83
short	directed	160	0.88	0.87	0.89
short	value-guided	40	0.84	0.82	0.85
short	value-guided	80	0.9	0.9	0.91
short	value-guided	160	0.93	0.93	0.94
long	directed	40	0.77	0.74	0.79
long	directed	80	0.85	0.84	0.87
long	directed	160	0.91	0.9	0.91
long	value-guided	40	0.84	0.83	0.85
long	value-guided	80	0.9	0.89	0.92
long	value-guided	160	0.94	0.93	0.94

Methods:

Second, participants played one practice round of the task which was followed by only 80 task rounds, as opposed to the originally proposed 160. This was driven by considerations for the total length of the experiment and participants' attention span. Prior to data collection, we conducted a power analysis based on publicly available data by Zaller et al.(Zaller et al., 2021) and found that the group-level effects in that dataset could be recovered 100% of the time when using a total of 80 games. We additionally tested the correlations between original and recovered parameter estimates for different numbers of games and found only minor differences between the versions for 80 and 180 games (r = 0.81 - 0.90 vs r = 0.88 - 0.94).

[...]

Discussion:

One limitation to our study is that we made several non-negligible modifications to the Horizon task when compared to its original implementation (Wilson, Geana, White, Ludvig, & Cohen, 2014). First, we reduced the total number of rounds from the original 160 rounds to 80 rounds. We verified through a priori power analyses that this reduced trial number would be sufficient to

detect expected effects. We additionally performed recovery analyses using simulated data of a varying numbers of rounds to ensure that in principle, reasonable parameter recovery could be expected from 80 rounds. The results from these simulations as well as a body of literature using versions of the Horizon task with reduced numbers of rounds(Dubois & Hauser, 2022b; Dubois et al., 2021; Zaller et al., 2021) encouraged us that 80 rounds should be a sufficient task length for the Horizon task. Second, in our implementation of the Horizon task participants were not provided with a history of their previous rewards, including the rewards received during the fixed choices. While this implementation has been successfully used before (Cogliati Dezza, Cleeremans, & Alexander, 2019), it adds a working memory load compared to the classic implementation, limiting comparability to previous studies. The choice to omit a reward history was motivated by considerations of comparability across tasks. Third, we altered the computational modeling of the Horizon task slightly to include two separate regression weights for value difference and information difference in a Bayesian Mixed Effects Regression as opposed to the general choice temperature applied to all predictors in the original implementation(Wilson et al., 2014). This choice was again motivated by the comparability between bandit tasks and follows a modeling approach suggested by Wilson et al.(Wilson, Bonawitz, Costa, & Ebitz, 2021a). Simulations showed the original model parameters to be highly correlated with the model parameters in our implementation (see Table S8 in the SI). While all these alterations were performed to improve the comparability between bandit tasks while maintaining a conceptual, even if not direct, comparability to previous studies of the Horizon task, it is possible that the inferior recoverability and reliability of the Horizon task could be driven by these limitations.

3.4 Comment 3: Additional details in the replicability and reliability sections

Point 3

In the Replicability section, the authors should provide the mean and 95% Highest Density Interval (HDI) values of the posterior distributions.
In the Reliability section, the authors should provide the ICC values. The authors also did not give descriptions for other task measures, such as Regret and p(optimal) that were shown in Figure 4A.

Response:

We would like to thank Reviewer 1 for this note and have added the requested details to the manuscript. Specifically, we inserted a table of estimates and upper and lower bounds of the HDIs. Further, we added the ICC values to the text in the reliability section. All values are also displayed in the figures. You can find the table and in-text additions below.

Task	Session	Strategy	Group-Effect	95% CI Lower	95% CI Upper
Horizon	Session 1	Value-guided	0.74	0.07	1.35
Horizon	Session 1	Directed	0.81	0.48	1.14
Horizon	Session 2	Value-guided	-0.07	-0.86	0.71
Horizon	Session 2	Directed	1.18	0.76	1.59
Two-armed	Session 1	Value-guided	-0.01	-0.01	0.00
Two-armed	Session 1	Directed	0.10	0.08	0.12
Two-armed	Session 1	Random	0.60	0.56	0.65
Two-armed	Session 2	Value-guided	0.03	0.02	0.04
Two-armed	Session 2	Directed	0.13	0.11	0.15
Two-armed	Session 2	Random	0.57	0.51	0.63
Restless	Session 1	Directed	-0.41	-0.62	-0.21
Restless	Session 1	Value-guided	0.15	0.14	0.16
Restless	Session 2	Directed	-0.47	-0.69	-0.26
Restless	Session 2	Value-guided	0.18	0.17	0.19

Table 2: Means as well as the upper and lower thresholds of the 95% highest density interval of the group-level posterior distributions (values rounded to two decimals).

In particular, the ICC3(C,1) of value-guided and directed exploration were 0.105 and 0.300, respectively, in the Horizon task; 0.471 and 0.513, respectively, in the Restless bandit; and 0.473 and 0.511, respectively, in the Two-armed bandit.

[...]

Besides the model parameters, we also analyzed a few model-free task-based measures. The task-based measures, including performance and switch probability, exhibited moderate to good reliability (ICC3(C,1) = 0.21 - 0.70). Remarkably, the switch probability (i.e. the proportion of trials on which participants selected a different arm than on the previous trial) was consistently more reliable than any of the model-based parameters (ICC3(C,1) = 0.56 - 0.69; see Figure 4A).

3.5 Comment 4: Errors in the reported values

Point 4

In the External & Criterion Validity section, there is an error in the statistical value presented in this sentence: "strongest correlation was .21 between value-guided exploration on the Restless bandit and the questionnaire measure of Openness to new experiences; see Figure 4E". However, in the correlation matrix in Figure 4E, the corresponding r value is 0.04. In addition, there is another error presented in this sentence: "However, we observed strong correlations between task-based measures and model parameters on the one hand and working memory capacity on the other (highest r = 0.4)." In the correlation matrix in Figure 4E, it seems like the highest r is 0.35. However, high r values pertain only to the measure "P(optimal) horizon." The rest of the values in the correlation matrix were equal or less than 0.18. Not sure whether it is sufficient to say, "This suggests that measures from the bandit paradigms are affected by individual differences in cognitive capacity".

Response:

We would like to thank Reviewer 1 for being so attentive and apologise for the oversight. We adjusted the paragraph according to the correct statistics. You can find the corrected paragraph below:

> We only observed relatively small correlations between the model parameters of exploration and questionnaire-based measures of exploration (strongest correlation was .18 between directed exploration in the horizon task and the questionnaire measures of cognitive and somatic anxiety; see Figure 4E). However, both task-based measures and parameter estimates showed small to moderate correlations with memory capacity (highest r = 0.35).

3.6 Comment 5: Additional information about exploration strategies

Point 5

In paragraph 3 of the Improving the Measurement of Exploration section, the authors only focus on data from the long horizon condition. The authors should discuss and clarify more regarding the change in the definitions. Namely, here directed exploration is defined as the information bonus parameter in horizon 6 while random exploration is defined as the inverse temperature in horizon 6. This is important since the labels in the figures remain the same throughout the paper, while the underlying definitions are not.

Response:

We would like to thank Reviewer 1 for this clarification. Generally, we moved the detailed explanations of the changes to the modeling procedure in the Methods section, to improve the flow of the paper. To

ensure that the change in underlying meaning of parameter estimate labels for the Horizon task is clear, we added the following sentences to the first paragraph of the section "Improving the Measurement of Exploration" in the Results section:

> Importantly, from here on out, the labels of directed and value-guided exploration in the Horizon task refer to only the estimates of these strategies in the long Horizon condition and not the difference in estimates between the long and the short Horizon.

3.7 Comment 6: Changed definition of exploration in the Horizon task

Point 6

In Figure 4, additional information about the acronyms should be provided in figure caption. For example, “Exploration” in Figure 4E corresponds to scores from the Curiosity and Exploration Inventory (CEI).

Response:

We thank Reviewer 1 for this note and added the following sentence to the caption of Figure 4.

> Note: Exploration refers to the CEI, Openness to the BIG5 Openness subscale, Cognitive and Somatic Anxiety to the respective STICSA subscales, Depression to the PHQ-9, and Negative and Positive mood to the respective PANAS subscales.

3.8 Comment 7: Missing information about latent factor modeling

Point 7

In paragraph 4 of the Improving the Measurement of Exploration section, the authors describe computing latent factors of the exploration strategies. However, the exact method for this analysis were not included in the Methods section or the supplementary information file.

Response:

We thank Reviewer 1 for pointing out this missing information. We added several paragraphs to the Methods section describing this procedure. You can also find those paragraphs here:

Structural equation modeling (SEM) is a latent-variable approach that attempts to estimate the scores of underlying latent constructs using a set of observable indicator variables by modeling the underlying variance-covariance matrix of all observed variables (Little, 2023). Before creating a structural model relating different constructs, it is recommended to create separate measurement models for every measured construct (Little, 2023) to test whether the individual constructs are actually measurable. One advantage of SEM compared to principle component analysis (i.e., PCA), is that methods variance, for example variance due task strategies or variance due to parameter trade-offs, can be modeled as correlated residuals in the indicator variables (Schmiedek, Oberauer, Wilhelm, Süß, & Wittmann, 2007). This approach ensures that variance in model parameters not due to the underlying construct is not propagated forward, for example in a correlation with another measure. We want to highlight here that the common practice of using model parameters from a single task propagates methods variance and error variance, and therefore produces biased correlation estimates. The propagation of methods variance also happens within a PCA approach. Model interpretation requires an acceptable fit of the model to the data. We refer to the guidelines provided by (McDonald & Ho, 2002), which state that an acceptable fit is reflected in a Comparative Fit Index (CFI) $> .9$, and a root mean squared error of approximation (RMSEA) $< .08$. Because value guided exploration traded off heavily with random exploration in the two-armed bandit, and because there were inconsistent relationships between the value-guided exploration parameters in the three tasks using the original models, we focused the latent modeling on the parameters from the improved models. We created measurement models for working memory capacity, value-guided exploration and directed exploration using the respective indicators (for working memory capacity, we used the proportion of responses reported in the correct serial position, first aggregated per set size and then aggregated over set sizes for each participant). These measurement models were just identifiable with df = 0 and the respective factor loadings are displayed in Figure 3.

We then created two structural models, one model with two latent exploration strategies (i.e., value-guided and directed), and one model with only one latent exploration strategy (i.e., all six indicators loading on the same construct). The former model did not converge using the data from the first session. We therefore fit the same two models to the data from the second session (CFI = 0.989, RMSEA = 0.045 for the two-strategy model, and CFI = 0.868, RMSEA = 0.186 for the one-strategy model). Approximating the Bayes factor between the two models with the method provided by (Wagenmakers, 2007) showed that the two-strategy model was clearly preferred.

Because directed exploration was not measurable in the first session, we fitted a structural model only including value-guided exploration and working memory capacity on the data from both sessions to calculate the reliability of the two factors. This model fitted well (CFI = 0.971, RMSEA = 0.059) and showed that both latent factors are measurable reliably over a time period of six weeks. The respective correlations were .78 and .91 for value-guided exploration and working memory capacity, respectively. Eventually, we fit a model including the two exploration strategies and working memory capacity, and the five questionnaire-based constructs (i.e., openness to new

experiences, exploration, positive mood, negative mood, and anxiety/depression) to the data from Session 2. Note that we computed the questionnaire constructs based on compound scores (i.e., not item-level scores) as has been validated and suggested before(Ree, French, MacLeod, & Locke, 2008; Kroenke, Spitzer, & Williams, 2001; Kashdan, Rose, & Fincham, 2004; Soto & John, 2017; Watson, Clark, & Tellegen, 1988). The fit of this joint model was acceptable with a CFI of 0.96 and an RMSEA of 0.05. The latent-level correlations based on this final model can be found in Figure 6.

3.9 Comment 8: Explain the pattern diverging from previously published studies

Point 8

In paragraph 5 of the Improving the Measurement of Exploration section, the authors stated that "Parameter recovery for the Horizon task, however, was on average unchanged." This again could be due to low number of data points (i.e., only data from the long horizon condition, N $<$ 80). The authors also stated, "It is visible in Fig. 5 though, that participants barely used directed exploration on average" and "Hence, the increase in directed exploration from the short to the long horizon, as reported before, reflects rather a decreased tendency to avoid uncertainty than active exploration." Additional explanation is needed to clarify this conclusion. For example, is it due to the observation that the group level information bonus parameter had values that were indistinguishable from zero? Note that details of the group level statistics were also not provided. Further, the increase in exploration in H6 compared to H1 on this task is well-established in quite a few studies. So it should be addressed why a different pattern is seen here.
In paragraph 3 of the Discussion section, the statements on uncertainty avoidance should be revised if Comment No. 9 is addressed.

Response:

We would like to thank Reviewer 1 for pointing out this important difference between our implementation of the Horizon task and the common implementations. We adjusted the phrasing in the Results section to clarify what is meant by saying that we do not find directed exploration in the long Horizon condition. We also added a table that contains the precise statistics for the group-level effects both before our adjustments to the modelling (mentioned above) and after (pasted below). Further, we added a paragraph to the discussion, outlining how the low number of datapoints could have affected the reliability of the Horizon task (mentioned earlier and pasted below again for your convenience).

Hence, the increase in directed exploration from the short to the long horizon, as reported before, reflects rather an avoidance of uncertainty in the short horizon that is attenuated in the long

Task	Session	Strategy	Group-Effect	95% CI Lower	95% CI Upper
Horizon	Session 1	Value-guided	3.62	3.23	4.04
Horizon	Session 1	Directed	0.32	0.06	0.57
Horizon	Session 2	Value-guided	4.79	4.23	5.37
Horizon	Session 2	Directed	-0.15	-0.45	0.13
Two-armed	Session 1	Value-guided	0.13	0.12	0.14
Two-armed	Session 1	Directed	0.15	0.13	0.18
Two-armed	Session 2	Value-guided	0.18	0.17	0.19
Two-armed	Session 2	Directed	0.17	0.15	0.20

Table 3: Group effects and confidence intervals for different strategies across sessions and tasks after improvements to the modeling (values rounded to two decimals).

Horizon as opposed to active uncertainty-seeking in the long Horizon.

Discussion:

One limitation to our study is that we made several non-negligible modifications to the Horizon task when compared to its original implementation (Wilson et al., 2014). First, we reduced the total number of rounds from the original 160 rounds to 80 rounds. We verified through a priori power analyses that this reduced trial number would be sufficient to detect expected effects. We additionally performed recovery analyses using simulated data of a varying numbers of rounds to ensure that in principle, reasonable parameter recovery could be expected from 80 rounds. The results from these simulations as well as a body of literature using versions of the Horizon task with reduced numbers of rounds(Dubois & Hauser, 2022b; Dubois et al., 2021; Zaller et al., 2021) encouraged us that 80 rounds should be a sufficient task length for the Horizon task. Second, in our implementation of the Horizon task participants were not provided with a history of their previous rewards, including the rewards received during the fixed choices. While this implementation has been successfully used before (Cogliati Dezza et al., 2019), it adds a working memory load compared to the classic implementation, limiting comparability to previous studies. The choice to omit a reward history was motivated by considerations of comparability across tasks. Third, we altered the computational modeling of the Horizon task slightly to include two separate regression weights for value difference and information difference in a Bayesian Mixed Effects Regression as opposed to the general choice temperature applied to all predictors in the original implementation(Wilson et al., 2014). This choice was again motivated by the comparability between bandit tasks and follows a modeling approach suggested by Wilson et al.(Wilson et al., 2021a). Simulations showed the original model parameters to be highly correlated with the model parameters in our implementation (see Table S8 in the SI). While all these alterations were performed to improve the comparability between bandit tasks while maintaining a conceptual, even if not direct, comparability to previous studies of the Horizon task, it is possible that the

inferior recoverability and reliability of the Horizon task could be driven by these limitations.

3.10 Comment 9: Different operationalization of the Horizon task compared to previously published studies

Point 9

Another reason giving rise to this pattern of results could be the way the Horizon task was operationalized in this paper, which is different from prior studies that utilized this task. In the Methods – Bandit tasks section, the authors stated, "We adapted the same general procedure as in the original publication with the sole difference that the history of rewards was not available to participants." This change differs from prior studies where forced-choice outcomes appear explicitly on screen (i.e., participants did not need to remember which option was chosen once or three times). This is not a trivial change, and it alters memory demands, making it less comparable to previous studies. This could explain why the information bonus parameter in the Horizon 6 condition of this task is well above 0 in prior studies using hierarchical Bayesian models for fitting. For example:

- Figure 5: Krueger, P. M., Wilson, R. C., & Cohen, J. D. (2017). Strategies for exploration in the domain of losses. Judgment and Decision Making, 12(2), 104-117.
- Figure 4: Zajkowski, W.K., Kossut, M., Wilson, R.C., 2017. A causal role for right frontopolar cortex in directed, but not random, exploration. Elife 6, e27430.

Response:

We would like to thank Reviewer 1 for pointing out this important difference between the original implementation and our version. We added a paragraph to the discussion section where we discuss how the difference in implementation could have affected the results. You can find it below:

One limitation to our study is that we made several non-negligible modifications to the Horizon task when compared to its original implementation (Wilson et al., 2014). First, we reduced the total number of rounds from the original 160 rounds to 80 rounds. We verified through a priori power analyses that this reduced trial number would be sufficient to detect expected effects. We additionally performed recovery analyses using simulated data of a varying numbers of rounds to ensure that in principle, reasonable parameter recovery could be expected from 80 rounds. The results from these simulations as well as a body of literature using versions of the Horizon task with reduced numbers of rounds(Dubois & Hauser, 2022b; Dubois et al., 2021; Zaller et al., 2021) encouraged us that 80 rounds should be a sufficient task length for the Horizon task.

Second, in our implementation of the Horizon task participants were not provided with a history of their previous rewards, including the rewards received during the fixed choices. While this implementation has been successfully used before (Cogliati Dezza et al., 2019), it adds a working memory load compared to the classic implementation, limiting comparability to previous studies. The choice to omit a reward history was motivated by considerations of comparability across tasks. Third, we altered the computational modeling of the Horizon task slightly to include two separate regression weights for value difference and information difference in a Bayesian Mixed Effects Regression as opposed to the general choice temperature applied to all predictors in the original implementation(Wilson et al., 2014). This choice was again motivated by the comparability between bandit tasks and follows a modeling approach suggested by Wilson et al.(Wilson et al., 2021a). Simulations showed the original model parameters to be highly correlated with the model parameters in our implementation (see Table S8 in the SI). While all these alterations were performed to improve the comparability between bandit tasks while maintaining a conceptual, even if not direct, comparability to previous studies of the Horizon task, it is possible that the inferior recoverability and reliability of the Horizon task could be driven by these limitations.

3.11 Comment 10: Model validation in the by-trial analysis

Point 10

In addition, the authors describe a further analysis based on the Two-armed bandit task before making the conclusion "A similar pattern was visible in the Two-armed bandit when analyzing choices for each trial separately." The figure caption in Figure S3 stated "Parameter values for value-guided and directed exploration plotted over trials when a hierarchical model is separately fitted to the choices in trials 2 - 10 in the Two-armed bandit task." This suggests for each trial only 30 data points (30 games) for each participant. Though the sample size is relatively large, N = 175, further model validation analyses seem necessary.

Response:

We would like to thank Reviewer 1 for pointing out this important missing information and added a Figure to the Supplementary Information that confirm the recoverability of the trial-by-trial models despite their limited number of trials. We included it below for your convenience:

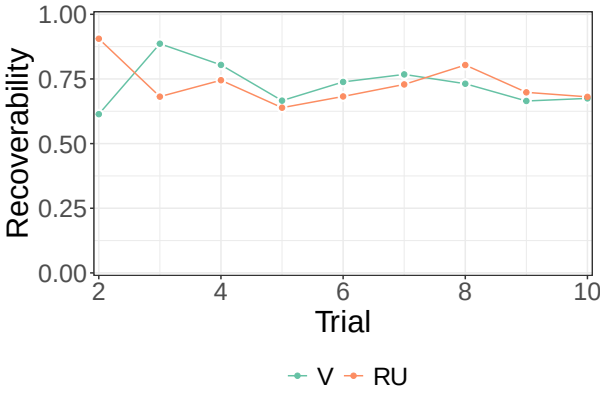


Figure 1: Recoverability of value-guided and directed exploration when analyzing choices 2-10 in the Two-armed bandit separately. On average, the MAP estimate for each participant, was recoverable with a correlation of .73 and .73 for value-guided exploration and directed exploration, respectively.

3.12 Comment 11: Add more details about the latent factor modeling

Point 11

In paragraph 2 of the Discussion section, the authors stated, "Following these improvements, test-retest reliability of single model parameters remained mediocre, but we successfully extracted a temporally stable latent trait of value-guided exploration, with a reliability of 0.91." This result was not mentioned in the Results section or in the figures that I could find. "Moreover, we identified a latent factor of directed exploration in the second session, but not in the first session, which may indicate that participants converged upon a strategy after sufficient task exposure." This was also not mentioned in the Results section.

Following Comment No. 8, there is no description of the methods used for the latent factor analysis or whether it was conducted separately for each session.

Response:

We would like to thank Reviewer 1 for pointing out this lack of details. We have added details on the reliability of the latent factor for value-guided exploration to the results section along with details on which latent model was fit to which dataset (first vs second session). As mentioned earlier, we also included additional details on the latent factor modeling in the methods sections. You can find the changes to the methods and to the results below for your convenience.

Please note that we slightly modified the latent factor modeling compared to the initial submission. Namely, to test the external and criterion validity following the improvements to the computational models, we estimated one joint structural equation model including all bandit tasks, all working memory tasks and all compound scores from the questionnaires. We included details on this computation in the methods section and you can also find them below. Importantly, this change lead to small changes in the reported correlations, albeit no changes in the qualitative conclusions.

Results:

Only the latent factor for value-guided exploration converged in both sessions, allowing us to assess the temporal stability of this factor, which was very high (r = 0.78). The model that includes both directed and value-guided exploration only converged for the data from session 2, not session 1, suggesting that directed exploration might only emerge after sufficient exposure to the task

Methods:

Structural equation modeling (SEM) is a latent-variable approach that attempts to estimate the scores of underlying latent constructs using a set of observable indicator variables by modeling the underlying variance-covariance matrix of all observed variables (Little, 2023). Before creating a structural model relating different constructs, it is recommended to create separate measurement models for every measured construct (Little, 2023) to test whether the individual constructs are actually measurable. One advantage of SEM compared to principle component analysis (i.e., PCA), is that methods variance, for example variance due task strategies or variance due to parameter trade-offs, can be modeled as correlated residuals in the indicator variables (Schmiedek et al., 2007). This approach ensures that variance in model parameters not due to the underlying construct is not propagated forward, for example in a correlation with another measure. We want to highlight here that the common practice of using model parameters from a single task propagates methods variance and error variance, and therefore produces biased correlation estimates. The propagation of methods variance also happens within a PCA approach.

Model interpretation requires an acceptable fit of the model to the data. We refer to the guidelines provided by (McDonald & Ho, 2002), which state that an acceptable fit is reflected in a Comparative Fit Index (CFI) $> .9$, and a root mean squared error of approximation (RMSEA) $< .08$. Because value guided exploration traded off heavily with random exploration in the two-armed bandit, and because there were inconsistent relationships between the value-guided exploration parameters in the three tasks using the original models, we focused the latent modeling on the parameters from the improved models. We created measurement models for working memory capacity, value-guided exploration and directed exploration using the respective indicators (for working memory capacity, we used the proportion of responses reported in the correct serial position, first aggregated per set size and then aggregated over set sizes for each participant). These measurement models were just identifiable with df = 0 and the respective factor loadings are displayed in Figure 3.

We then created two structural models, one model with two latent exploration strategies (i.e., value-guided and directed), and one model with only one latent exploration strategy (i.e., all six indicators loading on the same construct). The former model did not converge using the data from the first session. We therefore fit the same two models to the data from the second session (CFI = 0.989, RMSEA = 0.045 for the two-strategy model, and CFI = 0.868, RMSEA = 0.186 for the one-strategy model). Approximating the Bayes factor between the two models with the method provided by (Wagenmakers, 2007) showed that the two-strategy model was clearly preferred.

Because directed exploration was not measurable in the first session, we fitted a structural model

only including value-guided exploration and working memory capacity on the data from both sessions to calculate the reliability of the two factors. This model fitted well (CFI = 0.971, RMSEA = 0.059) and showed that both latent factors are measurable reliably over a time period of six weeks. The respective correlations were .78 and .91 for value-guided exploration and working memory capacity, respectively. Eventually, we fit a model including the two exploration strategies and working memory capacity, and the five questionnaire-based constructs (i.e., openness to new experiences, exploration, positive mood, negative mood, and anxiety/depression) to the data from Session 2. Note that we computed the questionnaire constructs based on compound scores (i.e., not item-level scores) as has been validated and suggested before(Ree et al., 2008; Kroenke et al., 2001; Kashdan et al., 2004; Soto & John, 2017; Watson et al., 1988). The fit of this joint model was acceptable with a CFI of 0.96 and an RMSEA of 0.05. The latent-level correlations based on this final model can be found in Figure 6.

3.13 Comment 12: Missing details about two-armed bandit task

Point 12

Note also that in the Methods – Bandit tasks section, the authors did not say whether the two options in the Two-Armed bandit task were explicitly labeled as "Stable" or "Fluctuating." Without these labels, as used in the original version, more cognitive effort would be required to keep track of not only the values of the options but also option types, which could potentially interact with the level of exploration. This is again fairly crucial, as changing the tasks makes them incomparable to findings in previous studies.

In the Methods – Bandit tasks section, following Comment No. 9, the authors should mention whether the options were explicitly labeled as "Stable" or "Fluctuating" as in prior studies using the task.

Together, these details need to be clarified to determine the validity of the concluding statement.

Response:

We would like to thank Reviewer 1 for this important clarification and added a statement to the Methods section. You can find the statement below:

Note that arms were not explicitly labeled as constant or drifting, which represents a change from Fan et al. 2023.

3.14 Comment 13: reference error

Point 13

There also appears to be an error in a reference in this paragraph, specifically, “Anvari et al.?”.

Response:

We would like to thank Reviewer 1 for pointing out this oversight. We have corrected it in the revised version of the manuscript.

3.15 Comment 14: Clarification on how information value was coded in the Horizon task

Point 14

In the Methods - Computational Modeling section, for the Horizon task, according to the paper, the information value for arm j on the first free choice (t = 5) is defined as follows:
$I_j(t = 5) = \Sigma_(t = 1)^4 \delta_j(t)$
Here, $\delta_j(t) = 1$ if arm j was chosen on trial t, and 0 otherwise. Following this definition, $[I_1(t = 5), I_2(t = 5)]$ could be [3,1], [1,3] or [2,2]. Thus, $I_1(t = 5) - I_2(t = 5)$ could either be 2, -2, or 0. However, in Wilson et al (2014), $I_1(t = 5) - I_2(t = 5)$ was instead either 1, -1 or 0.

Response:

We would like to thank the reviewers for this clarification. Since we fit the model as a Hierarchical Bayesian Mixed Effects Regression, we standardized the information difference to be either -0.5, 0.5 or 0, in an effort to ease the model fitting. The equation was merely written in this form to aid understanding. We added a line that outlines how the information difference, that we used in the logistic regression was computed. You can find the updated section below for your convenience.

> In this approach, exploration is defined as the difference in the model parameters between the long horizon and the short horizon conditions. For value-guided exploration, that is reflected in the difference between the parameters relating $E_1(t = 5) - E_2(t = 5)$ to choice probability. For directed exploration, it is reflected in the difference between the parameters relating $(I_1(t = 5) - I_2(t = 5))/4$ to choice probability (we divided the difference by 4 to create an indicator variable with levels -0.5, 0, 0.5 to aid with fitting in the logistic regression). For the Two-armed bandit, we used the model proposed by Gershman(Gershman, 2018) (with the sole difference that we used a logistic regression instead of a probit regression for convergence reasons), and for the Restless bandit, we used the model introduced by Speekenbrink and Konstantinidis(Speekenbrink & Konstantinidis, 2015)

3.16 Comment 15: Different definition of value-guided exploration than in previously published studies

Point 15

In addition, the authors stated, "For value-guided exploration, that is reflected in the difference between the parameters relating $E_1(t = 5) - E_2(t = 5)$ to choice probability. For directed exploration, it is reflected in the difference between the parameters relating $I_1(t = 5) - I_2(t = 5)$ to choice probability," and "The choice models in the Horizon task and in the Two-armed bandit were implemented via a logistic regression: $P(C(t) = j) = 1/(1 + exp(-(\beta_0 + \beta * x)))$ with x containing the independent variables $E_1(t) - E_2(t)$, $V_1(t) - V_2(t)$ (or $I_1(t) - I_2(t)$ in the case of the Horizon task), and β containing the respective parameters of the logistic regression. Note that β_0, the intercept of the regression, refers to a side bias, i.e., if someone preferentially chooses the left or the right arm without any further knowledge." However, the choice model in Wilson et al (2014) was instead:

$P(C(t) = a) = \frac{1}{(1+exp(((Q_b - Q_a))/\sigma_d))} = \frac{1}{(1+exp((R_b - R_a + \alpha(I_b - I_a) + B(s_b - s_a))/\sigma_d))}$

There are notable mathematical differences here beyond change in notation. Namely, the inverse temperature (β) or decision noise ($1/\sigma_d$) is weighted on the expected value differences, which include the sum of expected reward, expected information, and side bias. However, the choice model in this study instead has separate β weights for expected reward, expected information, and side bias.

Response:

We would like to thank Reviewer 1 for pointing out this important clarification. In our paper, we indeed slightly modified the computational modeling of the Horizon task to improve comparability to the other bandit tasks. We used a modeling approach based on what was suggested by (Wilson, Bonawitz, Costa, & Ebitz, 2021b). To verify that our modeling approach was similar to the original approach suggested by (Wilson et al., 2014) we simulated data using this original modeling approach and recovered the parameters using our modeling approach. We find the correlations to be high enough to hope for the parameters to be interpretable in equivalent ways. You can find the correlations below as well as in the updated SI. Nevertheless, we elaborate on this limitation in the discussion section. You can find the corresponding section below for your convenience.

> One limitation to our study is that we made several non-negligible modifications to the Horizon task when compared to its original implementation (Wilson et al., 2014). First, we reduced the total number of rounds from the original 160 rounds to 80 rounds. We verified through a priori power analyses that this reduced trial number would be sufficient to detect expected effects. We additionally performed recovery analyses using simulated data of a varying numbers of rounds to ensure that in principle, reasonable parameter recovery could be expected from 80 rounds. The results from these simulations as well as a body of literature using versions of the Horizon

Table 4: Correlations between original model parameters that we simulated from and recovered parameters using our implementation. Directed exploration = cor(information bonus, weight of difference in information), random exploration = cor(inverse choice temperature, weight of difference in means), side bias = cor(side bias, Intercept of regression)

Horizon	directed exploration	random exploration	side bias
short	0.75	0.82	0.86
long	0.81	0.82	0.86

task with reduced numbers of rounds(Dubois & Hauser, 2022b; Dubois et al., 2021; Zaller et al., 2021) encouraged us that 80 rounds should be a sufficient task length for the Horizon task. Second, in our implementation of the Horizon task participants were not provided with a history of their previous rewards, including the rewards received during the fixed choices. While this implementation has been successfully used before (Cogliati Dezza et al., 2019), it adds a working memory load compared to the classic implementation, limiting comparability to previous studies. The choice to omit a reward history was motivated by considerations of comparability across tasks. Third, we altered the computational modeling of the Horizon task slightly to include two separate regression weights for value difference and information difference in a Bayesian Mixed Effects Regression as opposed to the general choice temperature applied to all predictors in the original implementation(Wilson et al., 2014). This choice was again motivated by the comparability between bandit tasks and follows a modeling approach suggested by Wilson et al.(Wilson et al., 2021a). Simulations showed the original model parameters to be highly correlated with the model parameters in our implementation (see Table S8 in the SI). While all these alterations were performed to improve the comparability between bandit tasks while maintaining a conceptual, even if not direct, comparability to previous studies of the Horizon task, it is possible that the inferior recoverability and reliability of the Horizon task could be driven by these limitations.

3.17 Comment 16: Different model in the restless bandit task compared to the source study

Point 16

"The choice model for the restless bandit was an upper-confidence bound policy: $P(C(t) = j) = \frac{exp(\tau * E_j(t) + \beta * \sqrt{V_j(t) + \sigma^2_{innov} V_j(t) + \sigma^2_{innov})}}{\Sigma^4_{k=1} * exp(\tau * E_k(t) + \beta * \sqrt{V_k(t) + \sigma^2_{innov})}}$."
There is a redundant "*" in the denominator. The choice model in Daw et al (2006) was instead: "$P_{i,t} = \frac{exp(\beta[\mu^{pre}_{i,t} + \psi\sigma^{pre}_{i,t}])}{(\Sigma_j exp(\beta[\mu^{pre}_{j,t} + \psi\sigma^{pre}_{j,t}]))}$"
Here in the present study, again different from the original study, there were separate β weights for expected value and expected information.

Response:

We would like to thank Reviewer 1 for pointing out two important oversights. First, we removed the redundant "*" in the equation for the computational modeling of the Restless bandit. Second, we corrected the citation for the modeling, as we followed the more recent implementation by Speekenbrink & Konstantinidis (Speekenbrink & Konstantinidis, 2015) instead of the original implementation by Daw et al (Daw, O'Doherty, Dayan, Seymour, & Dolan, 2006). This implementation allowed us to tease apart the contributions of value and uncertainty, and therefore mirrored the separate treatment of value and uncertainty in our models of the Horizon task and the Two-armed bandit. That separate treatment would not be possible with the Daw et al. (2006) implementation, in which beta scales the sum of the value and the weighted uncertainty. You can find the corrected passage below:

> For the Two-armed bandit, we used the model proposed by Gershman(Gershman, 2018) (with the sole difference that we used a logistic regression instead of a probit regression for convergence reasons), and for the Restless bandit, we used the Kalman filter combined with the "softmax with exploration bonus" choice rule introduced by Speekenbrink and Konstantinidis(Speekenbrink & Konstantinidis, 2015)

3.18 Comment 17: Logistic regression instead of probit regression in the Two-armed bandit task

Point 17

"The choice models in the Horizon task and in the Two-armed bandit were implemented via a logistic regression: $P(C(t) = j) = 1/(1 + exp(-(\beta_0 + \beta * x)))$ with x containing the independent variables $E_1(t) - E_2(t)$, $V_1(t) - V_2(t)$ (or $I_1(t) - I_2(t)$ in the case of the Horizon task), and β containing the respective parameters of the logistic regression."

For the Two-armed bandit task, Gershman et al (2018) did not use a logistic regression. Instead, a probit regression was used to estimate choice probability:

"$P(a_t = 1|w) = \Phi(w_1 V + w_2 RU + w_3 \frac{V}{TU})$" where

"Estimated value difference (V): $Q_t(1) - Q_t(2)$.

Relative uncertainty (RU): $\sigma_t(1) - \sigma_t(2)$.

Total uncertainty (TU):$\sqrt{\sigma_t^2(1) + \sigma_t^2(2)}$"

Another difference is that relative uncertainty was defined as the difference in standard deviations where as $V_1(t) - V_2(t)$ in the present paper represents the difference in variance.

Response:

We would like to thank Reviewer 1 for this important clarification. Regarding the first point, the implementation of a probit regression was unfortunately not yielding convergence in the Hierarchical Bayesian framework, which is why we opted for a logistic regression instead. We have clarified this difference in the Methods section and pasted the changed section below for your convenience. Regarding the relative uncertainty, we implemented this exactly the way that it was described in Gershman et al (Gershman, 2018). However, it seems that the description in the manuscript is unclear on this point and we adjusted it accordingly. You can find that updated section below as well.

For the Two-armed bandit, we used the model proposed by Gershman(Gershman, 2018) (with the sole difference that we used a logistic regression instead of a probit regression for convergence reasons), and for the Restless bandit, we used the Kalman filter combined with the "softmax with exploration bonus" choice rule introduced by Speekenbrink and Konstantinidis(Speekenbrink & Konstantinidis, 2015).

...

The choice model for the restless bandit was an upper-confidence bound policy with a free inverse temperature parameter τ and a free directed exploration parameter β:

$$P(C(t) = j) = \frac{exp(\tau * E_j(t) + \beta * sqrt(V_j(t) + \sigma^2_{innov}))}{\sum_{k=1}^{4} exp(\tau * E_k(t) + \beta * sqrt(V_k(t) + \sigma^2_{innov}))} \quad (1)$$

The choice models in the Horizon task and in the Two-armed bandit were implemented via a logistic regression:

$$P(C(t) = j) = \frac{1}{1 + exp(-(\beta_0 + \boldsymbol{\beta} * \mathbf{x}))} \quad (2)$$

with $\mathbf{x}$ containing the independent variables $E_1(t) - E_2(t)$, $\sqrt{V_1(t)} - \sqrt{V_2(t)}$ (or $I_1(t) - I_2(t)$ in the case of the Horizon task), and $\boldsymbol{\beta}$ containing the respective parameters of the logistic regression. Note that β_0, the intercept of the regression, refers to a side bias, i.e., if someone preferentially chooses the left or the right arm without any further knowledge.

3.19 Comment 18: Explicitly state modeling changes

Point 18

In the Results-Improving the Measurement of Exploration section, the authors stated that they removed random exploration from the model in the Two-armed bandit task. In the Methods section, they should mention how the choice model was changed accordingly.

Response:

We would like to thank Reviewer 1 for this remark. The removal of random exploration from the model in the Two-armed bandit simply means the removal of one predictor in the logistic regression. Since the formulation of our choice model is general with respect to the names and number of predictors, it still applies after this change.

$$P(C(t) = j) = \frac{1}{1 + exp(-(\beta_0 + \boldsymbol{\beta} * \mathbf{x}))} \tag{3}$$

Based on the reviewer's remark, it seems like this point was not entirely clear and we clarified it some more in the Methods section. You can find the changed sentence below.

First, we removed random exploration from the model in the Two-armed bandit, thus reducing the number of predictors from three to two but leaving the rest of the model unchanged.

3.20 Comment 19: Name changes in modeling explicitly

Point 19

Given all of this, the first sentence "We used the same computational models as proposed in the source studies" does not appear accurate (similar statements mentioned elsewhere in the paper should also be revised, e.g., in paragraph 3 in the Results-General approach section). The authors should address the differences between their model adaptation and the models proposed in the original studies as well as the rationale behind these changes. In this section, the authors should also explain what the free parameters in these computational models were and explain how they relate to directed and value-driven/random exploration.

Response:

We would like to thank Reviewer 1 for pointing out the discrepancies between the original implementations and our implementations of the tasks and the computational modeling. Throughout the paper, we toned down the wording regarding whether we used the original implementations of the computational models or simply modeling that was based on original implementations. We also included several clarifications throughout the manuscript regarding differences in how value-guided and directed exploration are implemented across tasks (General Approach) and differences to the original task implementations (Methods - Bandits) and a detailed discussion regarding the changes made to the Horizon task. You can find the added sections below.

Results - General Approach

To estimate the extent to which participants' choices were guided by different exploration strategies, we fit those computational models to the choices in the three bandit paradigms based on what has been previously proposed in the literature(Wilson et al., 2014; Gershman, 2018; Speekenbrink & Konstantinidis, 2015).

...

Despite these conceptual similarities, the models differ slightly in how value-guided and directed exploration are implemented: In the Two-armed bandit, value-guided exploration and directed exploration are implemented as the regression weights of the difference in expected rewards and expected uncertainty, respectively. In the Horizon task, value-guided and directed exploration are implemented as the difference in these regression weights between the long and the short Horizon condition. In the Restless bandit, value-guided and directed exploration are implemented as the inverse temperature and uncertainty bonus in an upper-confidence bound policy, respectively.

...

Methods - Bandits

While the changes compared to the source studies in the Horizon task (i.e., no history) and in the two-armed bandit (no labeling of the arms) prohibited a direct replication of the original studies, the current task set represents a conceptual replication (Crandall & Sherman, 2016) of the source studies. That allowed us to test whether the previous observations are generalizable to slightly different task setups.

...

One limitation to our study is that we made several non-negligible modifications to the Horizon task when compared to its original implementation (Wilson et al., 2014). First, we reduced the total number of rounds from the original 160 rounds to 80 rounds. We verified through a priori power analyses that this reduced trial number would be sufficient to detect expected effects. We additionally performed recovery analyses using simulated data of a varying numbers of rounds to ensure that in principle, reasonable parameter recovery could be expected from 80 rounds. The results from these simulations as well as a body of literature using versions of the Horizon

task with reduced numbers of rounds(Dubois & Hauser, 2022b; Dubois et al., 2021; Zaller et al., 2021) encouraged us that 80 rounds should be a sufficient task length for the Horizon task. Second, in our implementation of the Horizon task participants were not provided with a history of their previous rewards, including the rewards received during the fixed choices. While this implementation has been successfully used before (Cogliati Dezza et al., 2019), it adds a working memory load compared to the classic implementation, limiting comparability to previous studies. The choice to omit a reward history was motivated by considerations of comparability across tasks. Third, we altered the computational modeling of the Horizon task slightly to include two separate regression weights for value difference and information difference in a Bayesian Mixed Effects Regression as opposed to the general choice temperature applied to all predictors in the original implementation(Wilson et al., 2014). This choice was again motivated by the comparability between bandit tasks and follows a modeling approach suggested by Wilson et al.(Wilson et al., 2021a). Simulations showed the original model parameters to be highly correlated with the model parameters in our implementation (see Table S8 in the SI). While all these alterations were performed to improve the comparability between bandit tasks while maintaining a conceptual, even if not direct, comparability to previous studies of the Horizon task, it is possible that the inferior recoverability and reliability of the Horizon task could be driven by these limitations.

3.21 Comment 20: Concluding remark

Point 20

As a general conclusion, the paper makes a nice contribution to a general literature on evaluating the utility and validity of exploration tasks. However, several choices were made without a stated rationale that may prevent results from being comparable to the previous studies they aim to build on. These choices will need to be justified and the limitations should be more thoroughly acknowledged and discussed before this paper could be published. I hope the authors find these comments helpful in improving their paper.

Response:

We would like to thank Reviewer 1 again for all their helpful suggestions. We are confident that the changes we made significantly improved the quality of the paper and hope that the reviewer agrees with the adjustments we made.

4 Reviewer 1

5 Authors' Response to Reviewer 2

5.1 General remarks

General Comments.

Witte et al. explore the psychometric properties of three bandit tasks and models from previous literature: the Two-armed bandit, Restless bandit, and Horizon task. They find poor to moderate reliability, weak task convergence, and no significant associations with self-reports or psychiatric symptoms. By simplifying models, they were able to achieve slightly better convergent validity among the parameter measures across the tasks, but divergent validity, criterion validity, and external validity were still non-existent. The study is timely and contributes valuable insights into measurement challenges in computational psychiatry, but improving the clarity of the presentation, adding additional analyses, and providing broader contextualization could significantly improve the manuscript.

Response:

We would like to thank Reviewer 2 for their favorable evaluation of our study and their concise summary of our results. Below, we will address their detailed comments point-by-point.

5.2 Comment Major 1: Structure of the manuscript

Point 1

The manuscript's structure needs refinement

a. The introduction includes excessive result summaries. Reframe this to outline the planned analyses, reserving results for the appropriate sections

b. The "General approach" (Results) and "Improving the Measurement of Exploration" subsections contain substantial methodological details (e.g., model updates, CFA). Relocate these to the Methods section, as the current Methods lack any description of updated models or latent factor analysis.

Response:

We would like the thank Reviewer 2 for this helpful comment. We have restructured the paper and removed a large portion of the results from the introduction section. We also moved details on the computational modeling, model improvements and CFA to the methods sections. Lastly, we extended the descriptions of the latent factor analysis and changes to the computational modeling in the Methods section. You can find the added sections from the Methods below for your convenience.

First, we removed random exploration from the model in the Two-armed bandit, thus reducing the number of predictors from three to two but leaving the rest of the model unchanged. This change was motivated by the redundancy in the two strategies as indicated by the high correlation between their respective predictors (on average between .83 to .94 across trials 2-10 in the four conditions; see Fig. S3 in the SI)(Daoud, 2017), between the recovered parameters (-.79 and -.78; see Fig. 2), and between the estimates of the two strategies (-.88; see Fig. 4). In addition, random exploration in the Two-armed bandit likely captured value-guided variance given its positive correlation with value-guided exploration in the Restless bandit.

Second, we only used the data from the long horizon condition to estimate value-guided exploration and directed exploration in the Horizon task. Traditionally, exploration is defined as the change in the respective regression parameter from the short to the long horizon condition. This is problematic because it differs from the definition as the absolute value of the respective parameters in the other two bandit tasks. Difference measures are moreover known to have a stronger contamination by error variance (Hedge, Powell, & Sumner, 2018; Enkavi et al., 2019; Salthouse & Hedden, 2002), which reduces their reliability. As it has been hypothesized and observed that people explore more in the long horizon condition than in the short horizon condition (Wilson et al., 2014; Dubois et al., 2021; Dubois & Hauser, 2022a), we exclusively focused on the first free choice from the former.

Third, we computed latent factors of the two remaining exploration strategies to remove unwanted sources of variance (i.e., task-specific variance, error variance, and variance due to parameter trade-offs induced during model fitting(Ratcliff & Tuerlinckx, 2002)) from the strategy estimates. For that we used a confirmatory factor analysis including correlations between residuals of parameters estimated from the same task to capture parameter trade-offs separately in the model (Schmiedek et al., 2007). That is, we explicitly tested whether the model parameters in the three tasks measured generalizable aspects of the two strategies. You can find more details on the Latent Factor Modeling in the following.

Latent Factor Modeling

Structural equation modeling (SEM) is a latent-variable approach that attempts to estimate the scores of underlying latent constructs using a set of observable indicator variables by modeling the underlying variance-covariance matrix of all observed variables (Little, 2023). Before creating a structural model relating different constructs, it is recommended to create separate measurement models for every measured construct (Little, 2023) to test whether the individual constructs are actually measurable. One advantage of SEM compared to principle component analysis (i.e., PCA), is that methods variance, for example variance due task strategies or variance due to parameter trade-offs, can be modeled as correlated residuals in the indicator variables (Schmiedek et al., 2007). This approach ensures that variance in model parameters not due to the underlying construct is not propagated forward, for example in a correlation with another measure. We want to highlight

here that the common practice of using model parameters from a single task propagates methods variance and error variance, and therefore produces biased correlation estimates. The propagation of methods variance also happens within a PCA approach.

Model interpretation requires an acceptable fit of the model to the data. We refer to the guidelines provided by (McDonald & Ho, 2002), which state that an acceptable fit is reflected in a Comparative Fit Index (CFI) $> .9$, and a root mean squared error of approximation (RMSEA) $< .08$. Because value guided exploration traded off heavily with random exploration in the two-armed bandit, and because there were inconsistent relationships between the value-guided exploration parameters in the three tasks using the original models, we focused the latent modeling on the parameters from the improved models. We created measurement models for working memory capacity, value-guided exploration and directed exploration using the respective indicators (for working memory capacity, we used the proportion of responses reported in the correct serial position, first aggregated per set size and then aggregated over set sizes for each participant). These measurement models were just identifiable with df = 0 and the respective factor loadings are displayed in Figure 3 .

We then created two structural models, one model with two latent exploration strategies (i.e., value-guided and directed), and one model with only one latent exploration strategy (i.e., all six indicators loading on the same construct). The former model did not converge using the data from the first session. We therefore fit the same two models to the data from the second session (CFI = 0.989, RMSEA = 0.045 for the two-strategy model, and CFI = 0.868, RMSEA = 0.186 for the one-strategy model). Approximating the Bayes factor between the two models with the method provided by (Wagenmakers, 2007) showed that the two-strategy model was clearly preferred.

Because directed exploration was not measurable in the first session, we fitted a structural model only including value-guided exploration and working memory capacity on the data from both sessions to calculate the reliability of the two factors. This model fitted well (CFI = 0.971, RMSEA = 0.059) and showed that both latent factors are measurable reliably over a time period of six weeks. The respective correlations were .78 and .91 for value-guided exploration and working memory capacity, respectively. Eventually, we fit a model including the two exploration strategies and working memory capacity, and the five questionnaire-based constructs (i.e., openness to new experiences, exploration, positive mood, negative mood, and anxiety/depression) to the data from Session 2. Note that we computed the questionnaire constructs based on compound scores (i.e., not item-level scores) as has been validated and suggested before(Ree et al., 2008; Kroenke et al., 2001; Kashdan et al., 2004; Soto & John, 2017; Watson et al., 1988). The fit of this joint model was acceptable with a CFI of 0.96 and an RMSEA of 0.05. The latent-level correlations based on this final model can be found in Figure 6.

5.3 Comment Major 2: Connect study to broader discussion in computational psychiatry

Point 2

The psychometric challenges detailed in the paper are not unique to the explore-exploit mechanisms and the paper would benefit from connecting this to the broader discussion introduced by Karvelis, P., Paulus, M. P., & Diaconescu, A. O. (2023). Individual differences in computational psychiatry: A review of current challenges. Neuroscience and Biobehavioral Reviews, 148, 105137.

Response:

We would like to thank Reviewer 2 for bringing to our attention this very relevant review. We reference it now in the introduction, results and the discussion section. You can find the relevant sections below.

Introduction:

However, we only know very little about the psychometric properties of computational measures of cognitive constructs in general (see Karvelis et al.(Karvelis, Paulus, & Diaconescu, 2023) for an overview and concrete methodological recommendations) and about exploration strategies as inferred using computational model parameters in particular.

Results:

We implemented all three models as hierarchical Bayesian models (following recommendations by Karvelis et al.(Karvelis et al., 2023)), which made parameter estimation more robust, improved out-of-sample prediction and the parameters' reliabilities, and did not change anything about the individual estimates and group-level patterns (see Tables S3 and S4 and Figures S1 and S2 in the SI).

Discussion:

In line with recent concerns about the psychometric limitations of computational tasks in psychiatry(Karvelis et al., 2023), we observed mixed results: While we observed good parameter recovery, there were substantial parameter trade-offs and weak signatures of directed exploration.

5.4 Comment Major 3: Reliability clarification

Point 3

The reliability warrants deeper exploration

a. Are practice effects evident in parameter estimates or task measures between sessions? The Horizon task's shift (value-guided exploration disappearing in session 2) suggests this. Reporting such effects would be valuable.

b. The choice of ICC(C,1) for consistency is unclear—why not ICC(A,1) for absolute agreement, which captures systematic differences like practice effects? Comparing both could reveal their impact on reliability.

c. Plotting parameter recovery alongside test-retest reliability (e.g., adding recovery results to Figure 4A) could clarify low reliability causes, as recovery sets an upper bound (see Karvelis et al., 2023).

d. With recovery in Figure 4A, Figure 2 becomes redundant—its off-diagonal correlations add little beyond Figure 4D.

e. This approach (c, d) applies to Figures 5 and 6 as well.

f. Test-retest reliability of working memory measures is unreported. What does prior literature (e.g., Oswald et al., 2015) say about their stability? Knowing reliability of all measures is key for interpreting validity findings.

g. Similarly, what about the reliability of self-report measures across sessions?

Response:

We would like to thank Reviewer 2 for these important clarification points and will address them one by one:

a) We did find small practice effects when comparing the proportion of optimal choices across sessions, but they only reached significance in the Horizon task and the Two-armed bandit. We added these findings to the Reliability section of the results. You can also find the added text below. While these practice effects could be a reason for value-guided exploration disappearing in session 2 in the original implementation, this seems unlikely: The disappearance of value-guided exploration in the original implementation merely shows that people did not differ in how much they sought out rewards in the long and the short Horizon condition. While it is possible that this would be influenced by practice, we are unsure why practice would effect the short Horizon so much more heavily than the long Horizon, thereby closing the gap between the two conditions.

b) We chose the ICC(C,1) for consistency instead of the ICC(A,1) for agreement, because we were mostly interested in the relative amount of exploration irrespective of practice effects. That is, although practice effects may exist in few-armed bandit tasks, our focus was on whether participants who explore more in the first session still explore more in the second session, as these individual differences were the main focus of our paper. Nevertheless, we repeated the reliability analyses using the ICC(A,1) and found very similar results. You can find the updated paragraph as well as a table of all ICC(C,1) and ICC(A,1) values below.

c+e) We added brackets to our reliability plot in Figures 4A and 6A and find them a very beneficial addition to the visualization. You can find the updated figures below.

d+e) We feel like the off-diagonal correlations are relevant enough to be demonstrated in this fashion to make them more instantly apparent to the reader. We therefore obted to retain that figure.

f+g) We added a paragraph to the results section outlining the excellent reliabilities of the working memory measures and self-report questionnaires. You can find that paragraph below.

Specifically, we focused on the consistency definition of the ICC(Shrout & Fleiss, 1979), because we were mostly interested in the relative amount of exploration irrespective of practice effects. That is, although practice effects may exist in few-armed bandit tasks, our focus was on whether participants who explore more in the first session still explore more in the second session. For the sake of completeness, we also report the results using the absolute agreement definition. In line with previous work on the Two-armed bandit (Schurr, Reznik, Hillman, Bhui, & Gershman, 2024), the model parameters demonstrated mediocre reliabilities at best, with some parameters having a poor reliability (Koo & Li, 2016). In particular, the ICC3(C,1) of value-guided and directed exploration were 0.105 and 0.300, respectively, in the Horizon task; 0.471 and 0.513, respectively, in the Restless bandit; and 0.473 and 0.511, respectively, in the Two-armed bandit. The results looked similarly when considering the ICC3(A,1), which also considers the average difference in a measure between sessions as a critical source of variability (e.g., practice effects, see Table S5 in the SI). Two exceptions were value-guided exploration and p(optimal) in the Two-armed bandit task. The drop from the ICC3(C,1) to the ICC3(A,1) may reflect the fact that participants generally responded more deterministically in the second session than in the first session. The shared variance between measurement time points (R^2) was less than 0.36 for all parameters, and for most even less than 0.27. The low reliability observed in the Horizon task may be partially attributed to the fact that exploration is defined by the difference between two experimental conditions, and difference scores are notoriously known for providing low reliabilities (Salthouse & Hedden, 2002; Hedge et al., 2018; Zorowitz & Niv, 2023). Besides the model parameters, we also analyzed a few model-free task-based measures. The task-based measures, including performance and switch probability, exhibited moderate to good reliability (ICC3(C,1) = 0.21 - 0.70). Remarkably, the switch probability (i.e. the proportion of trials on which participants selected a different arm than on the previous trial) was consistently more reliable than any of the model-based parameters (ICC3(C,1) = 0.56 - 0.69; see Figure 2A). We also tested whether there were any practice effects on the proportion of optimal choices in the bandit tasks, and found very small but statistically significant practice effects for the Horizon task and the Two-armed bandit but not the Restless bandit ($t(174) = 2.63, p = .009$, mean diff. $= 0.01$; $t(174) = 6.46, p < .001$, mean diff. $= 0.02$; $t(174) = 1.96, p = .051$, mean diff. $= 0.02$, respectively)

...

In contrast to the measures of the bandit tasks, retest reliability was good (i.e., ICC3(C,1) > 0.75)

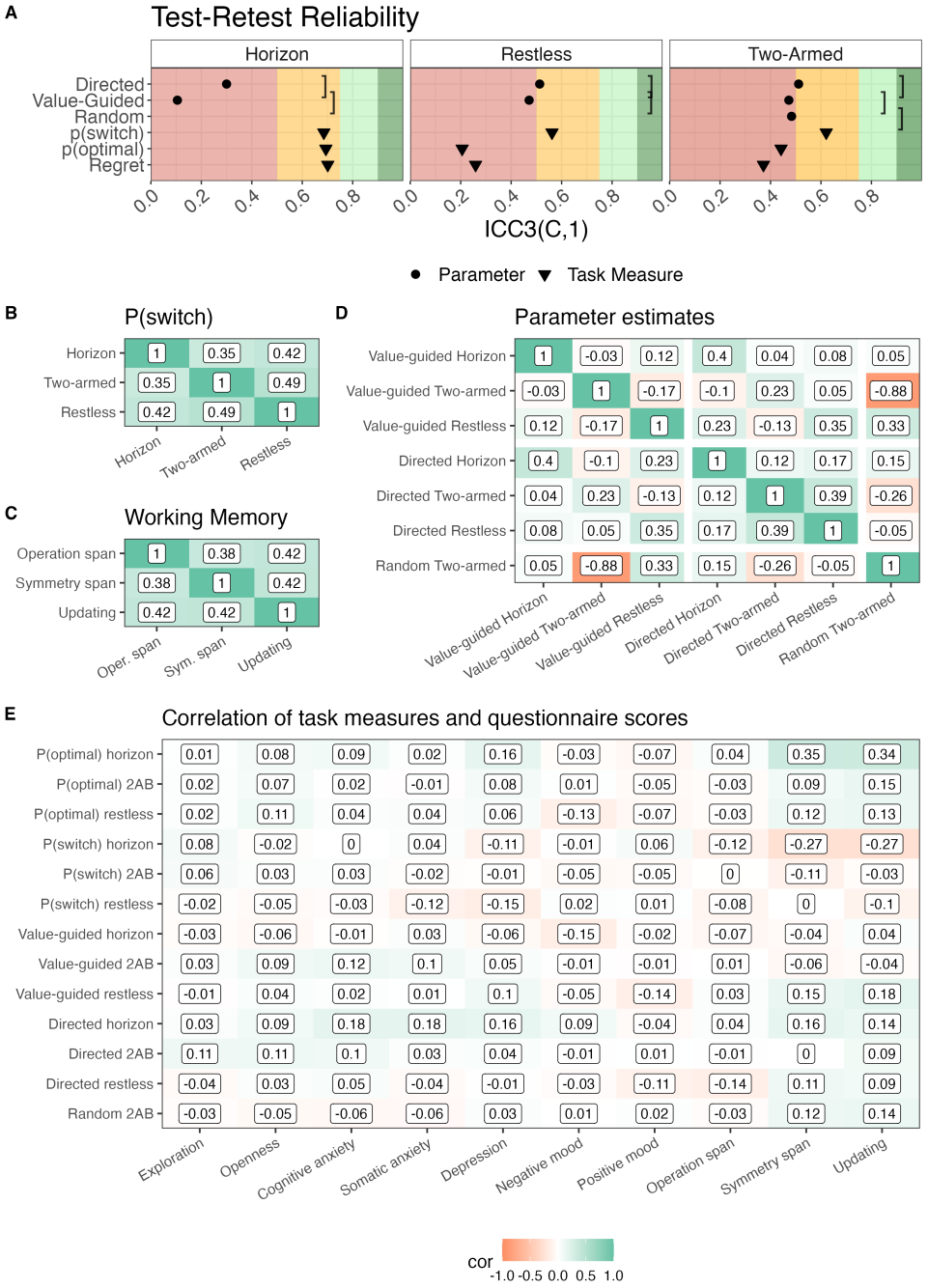


Figure 2: **A:** Test-retest reliability of model parameters and task measures. The red, orange, light green and dark green background indicates bad, acceptable, good and very good reliability, respectively. Circles indicate the reliabilities of model-based parameters. Crosses indicate the reliabilities of model-free task measures. Leftwards facing square brackets represent the recoverability of the parameters. **B:** Convergent validity of the trial-by-trial switch probability. Correlations between participants' average proportion of trials where they selected a different option than on the previous trial across tasks. **C:** Convergent validity of the recall accuracy in the working memory tasks. **D:** Convergent validity of the model parameters in the bandit tasks. Correlations between subject-level model parameters across tasks. **E:** External validity of task measures and model parameters. Correlations of subject-level task measures as well as model parameters with all questionnaire scores and performance in all three working memory tasks. Note: Exploration refers to the CEI, Openness to the BIG5 Openness subscale, Cognitive and Somatic Anxiety to the respective STICSA subscales, Depression to the PHQ-9, and Negative and Positive mood to the respective PANAS subscales.

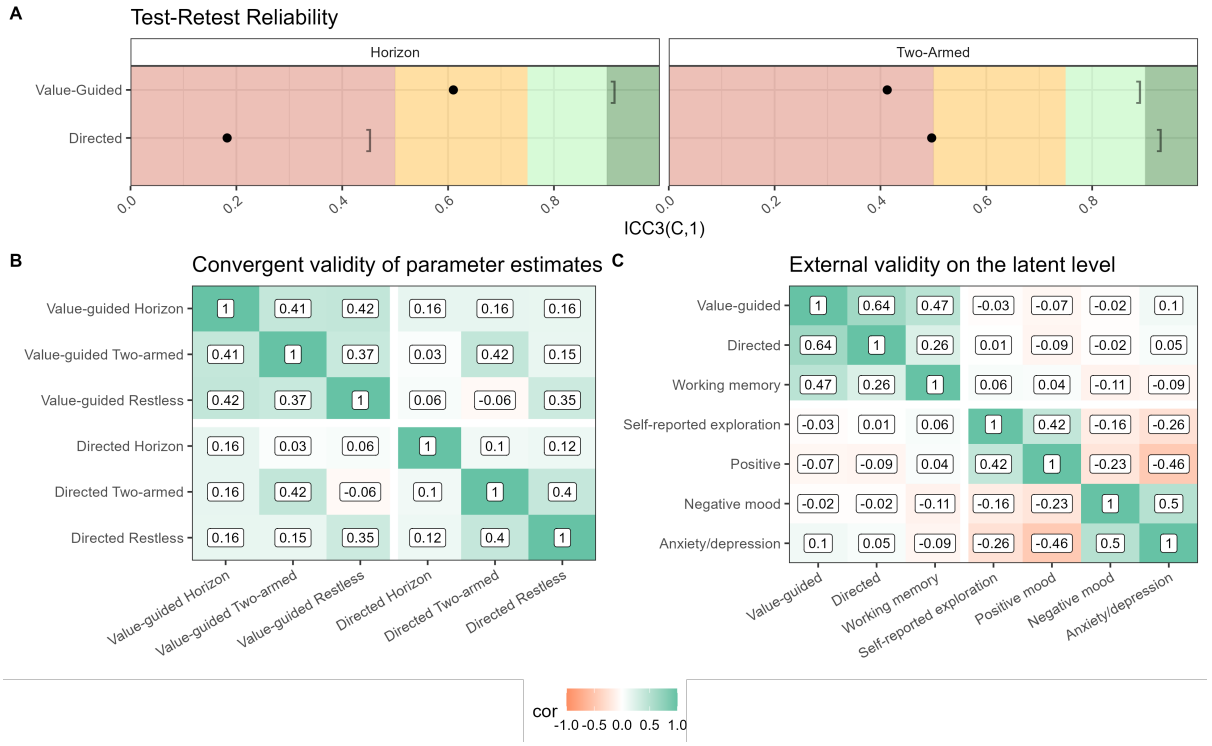


Figure 3: **A:** Test-retest reliability of model parameters from the Horizon task and the Two-armed bandit following our improvements to the modeling process. Red, orange, light green and dark green backgrounds indicate poor, acceptable, good and very good reliability, respectively. We again calculated the reliabilities in the form the respective intra-class correlation coefficient ICC(3,1)(Shrout & Fleiss, 1979) as a measure of consistency(McGraw & Wong, 1996)**B:** Convergent validity of the model parameters following improvements on the model fitting process. **C:** Correlations of latent factors from bandit tasks and questionnaires.

Task	**Parameter**	**ICC**	**Measure**	**Retest Reliability**
Horizon	Directed	Agreement	Parameter	0.273
Horizon	Value-Guided	Agreement	Parameter	0.0940
Horizon	p(switch)	Agreement	Task Measure	0.686
Horizon	p(optimal)	Agreement	Task Measure	0.694
Horizon	Regret	Agreement	Task Measure	0.701
Horizon	Directed	Consistency	Parameter	0.300
Horizon	Value-Guided	Consistency	Parameter	0.105
Horizon	p(switch)	Consistency	Task Measure	0.686
Horizon	p(optimal)	Consistency	Task Measure	0.694
Horizon	Regret	Consistency	Task Measure	0.701
Restless	Directed	Agreement	Parameter	0.514
Restless	Value-Guided	Agreement	Parameter	0.442
Restless	p(switch)	Agreement	Task Measure	0.557
Restless	p(optimal)	Agreement	Task Measure	0.203
Restless	Regret	Agreement	Task Measure	0.250
Restless	Directed	Consistency	Parameter	0.513
Restless	Value-Guided	Consistency	Parameter	0.471
Restless	p(switch)	Consistency	Task Measure	0.562
Restless	p(optimal)	Consistency	Task Measure	0.206
Restless	Regret	Consistency	Task Measure	0.259
Two-Armed	Directed	Agreement	Parameter	0.505
Two-Armed	Value-Guided	Agreement	Parameter	0.357
Two-Armed	Random	Agreement	Parameter	0.477
Two-Armed	p(switch)	Agreement	Task Measure	0.613
Two-Armed	p(optimal)	Agreement	Task Measure	0.391
Two-Armed	Regret	Agreement	Task Measure	0.370
Two-Armed	Directed	Consistency	Parameter	0.511
Two-Armed	Value-Guided	Consistency	Parameter	0.473
Two-Armed	Random	Consistency	Parameter	0.484
Two-Armed	p(switch)	Consistency	Task Measure	0.621
Two-Armed	p(optimal)	Consistency	Task Measure	0.442
Two-Armed	Regret	Consistency	Task Measure	0.372

Table 5: Retest reliability in the form of ICC3(C,1) and ICC3(A,1) for the model parameters of the original models and the task measures in the three tasks.

for the questionnaire scales and for the working memory measures. Two exceptions were self-reported exploration tendencies (ICC3(C,1) = 0.736) and working memory updating (ICC3(C,1) = 0.723), which, however, still had moderate reliabilities. We expected these results since the questionnaire scales have been validated before (Kashdan et al., 2004; Soto & John, 2017; Ree et al., 2008; Kroenke et al., 2001; Watson et al., 1988), and several studies have reported good reliability of the used measures in our working memory tasks (e.g., (Oswald, McAbee, Redick, & Hambrick, 2015; von Bastian, Souza, & Gade, 2016; Wilhelm, Hildebrandt, & Oberauer, 2013)).

5.5 Comment Major 4: Correction for multiple comparisons

Point 4

Multiple statistical tests are conducted. Addressing correction for multiple comparisons and the risk of overfitting (e.g., post-hoc model construction) would strengthen the analysis.

Response:

We would like to thank Reviewer 2 for bringing up this important issue. While we generally agree with the importance of accounting for multiple comparisons, we feel that this is a considerably less straight-forward necessity in the specific case of our paper: Especially in the first subsections of the results section, our focus is not on statistical significance, but rather on providing estimates of recoverability of model parameters, replicability of experimental patterns, and reliability of individual differences. However, in the case of the correlations between the latent variables (Figure 6C), correcting for multiple comparisons may be informative. Here, we perform a total of 21 correlation tests and report statistical significance for three of them (and null results for the majority of others). When applying a Bonferroni correction assuming 21 statistical tests, an appropriate cut-off for the p-value would be .002, which is well-above the p-values we report. The more important pattern in this section, however, is that there are no significant correlations between behavioral measures and questionnaire measures. Reducing the significance threshold for p-values due to multiple comparisons would not change anything about this conclusion, because it would only make these comparison even more insignificant. We therefore added the following sentence in the manuscript:

The two latent exploration factors were however highly correlated with each other ($r = 0.64$, $p < .001$) and with the latent working memory capacity factor ($r = 0.16$, $p = .004$ and $r = 0.37$, $p < .001$ for directed and value-guided exploration, respectively; note that the correlations between the latter three relationships remain significant when using Bonferroni's correction for multiple

comparisons).

5.6 Comment Major 5: No CFA on parameters of originally proposed models

Point 5

Why couldn't theoretically-informed latent constructs be computed with the original models (p. 2)? More explanation here would help readers understand the initial CFA failure.

Response:

We would like to thank Reviewer 2 for requesting this important clarification. Two reasons led us to omit the computation of the latent variables using the parameters of the original models. First, value guided exploration traded off heavily with random exploration in the two-armed bandit (i.e., -.88). Given this high correlation, we have to conclude that the Two-armed bandit does not allow us to make separate statements about individual differences in value-guided exploration and random exploration, because they are literally the same. Second, the relationships between the value-guided exploration parameters in the three tasks using the original models were inconsistent (with two correlations even below 0). For these two reasons, calculating a latent factor of value-guided exploration, using the original implementations of the models, is a priori guaranteed to fail. Therefore, we skipped the theoretically informed latent factor modeling with the parameters from the original models.

Because value guided exploration traded off heavily with random exploration in the two-armed bandit, and because there were inconsistent relationships between the value-guided exploration parameters in the three tasks using the original models, we focused the latent modeling on the parameters from the improved models.

5.7 Comment Major 6: Discuss validity of Exploration Strategies

Point 6

Given the stark findings (unstable factors, no external or criterion validity), discuss broader implications and promising directions. As you demonstrate, incremental model tweaks may help, but if core validity issues persist, what's the value of these tasks and models? Highlighting this challenge, even without solutions, would underscore the significance of your findings.

Response:

We would like to thank Reviewer 2 for this important remark. We added a paragraph to the discussion section summarizing the core issues we uncovered and the challenges they pose to the field more generally. You can find it below.

In conclusion, the lack of robust external and criterion validity observed in our study raises important questions regarding the interpretability and practical relevance of model-based exploration measures. While we were able to extract stable latent constructs—particularly for value-guided exploration—these constructs were not predictive of self-reported exploration tendencies, mood, or psychiatric symptoms. This disconnect suggests that, even with improved modeling and latent variable techniques, the external validity of these paradigms remains uncertain. If such tasks fail to capture variance that generalizes to real-world behaviors or clinically relevant constructs, their value as tools for individual differences research may be limited. At a minimum, these findings call for a reevaluation of the assumptions underlying current task designs and their intended applications. Future work may benefit from developing tasks that more strongly decouple exploration from performance demands, possibly incorporating richer, ecologically grounded scenarios, or explicitly modeling domain-specific forms of exploration.

5.8 Comment Minor 1

Point 7

"However, we observed strong correlations between task-based measures and model parameters on the one hand and working memory capacity on the other (highest r = 0.4)" – correlation of 0.4 or less probably does not deserve "strong" label; "moderate" would be more accurate. Also, I do not see 0.4 anywhere in fig 4e. 0.35 seems to be the highest? Also, it would be clearer to say "both task-based measures and parameter estimates showed moderate correlations with memory capacity" instead of the current cumbersome sentence structure.

Response:

We would like to thank Reviewer 2 for spotting this oversight. You can find the updated sentence below.

However, both task-based measures and parameter estimates showed small to moderate correlations with memory capacity (highest r = 0.35).

5.9 Comment Minor 2

Point 8

In your methods, you say "These two adaptations neither changed the pattern of group-level effects nor the absolute parameter values of the individual effects as compared to the originally proposed models" and "comparison showed that neither the group-level nor the subject-level parameter estimates were changed in our hierarchical Bayesian implementation compared to the originally proposed subject-level implementation" which is misleading because the whole point of the hierarchical estimation is to improve the subject-level estimates, which is what you see in the shrinkage effects.

Response:

We would like to thank Reviewer 2 for spotting this inaccuracy in our description. We rectified it and elaborated a bit more on how the changes to the modeling impacted the results. You can find the updated section below.

> These two adaptations did not change the pattern of group-level effects. The new implementation, however, led to three desired effects. First, it improved the parameter values of the individual effects as compared to the originally proposed models via shrinkage. In particular, it reduced the effect of parameter outliers when fitting individual participant data using Maximum Likelihood estimation, likely due to parameter trade-offs (see Figure S1 in the SI). Second, the model parameters had a higher reliability. Third, it improved the models' predictive accuracy on held-out data (see Figure S1 in the SI).

5.10 Comment Minor 3

Point 9

"Unifying the computational models to estimate the strategies.." not clear what you mean by unifying here

Response:

We would like to thank Reviewer 2 for requesting this important clarification. Based on this comment and other comments from Reviewer 1, we decided to remove this phrasing from the Results section entirely and instead added a longer section to the Methods where we detail the changes made to the computational modeling of the three tasks. You can find that section below.

Improved Models to Measure Exploration Strategies

First, we removed random exploration from the model in the Two-armed bandit, thus reducing the number of predictors from three to two but leaving the rest of the model unchanged. This change was motivated by the redundancy in the two strategies as indicated by the high correlation between their respective predictors (on average between .83 to .94 across trials 2-10 in the four conditions; see Fig. S3 in the SI)(Daoud, 2017), between the recovered parameters (-.79 and -.78; see Fig. 2), and between the estimates of the two strategies (-.88; see Fig. 4). In addition, random exploration in the Two-armed bandit likely captured value-guided variance given its positive correlation with value-guided exploration in the Restless bandit.

Second, we only used the data from the long horizon condition to estimate value-guided exploration and directed exploration in the Horizon task. Traditionally, exploration is defined as the change in the respective regression parameter from the short to the long horizon condition. This is problematic because it differs from the definition as the absolute value of the respective parameters in the other two bandit tasks. Difference measures are moreover known to have a stronger contamination by error variance (Hedge et al., 2018; Enkavi et al., 2019; Salthouse & Hedden, 2002), which reduces their reliability. As it has been hypothesized and observed that people explore more in the long horizon condition than in the short horizon condition (Wilson et al., 2014; Dubois et al., 2021; Dubois & Hauser, 2022a), we exclusively focused on the first free choice from the former.

Third, we computed latent factors of the two remaining exploration strategies to remove unwanted sources of variance (i.e., task-specific variance, error variance, and variance due to parameter trade-offs induced during model fitting(Ratcliff & Tuerlinckx, 2002)) from the strategy estimates. For that we used a confirmatory factor analysis including correlations between residuals of parameters estimated from the same task to capture parameter trade-offs separately in the model (Schmiedek et al., 2007). That is, we explicitly tested whether the model parameters in the three tasks measured generalizable aspects of the two strategies. You can find more details on the Latent Factor Modeling in the following.

5.11 Comment Minor 4

Point 10

Figures S1 and S2. Indicate in the caption which tasks you are talking about.

Response:

We would like to thank Reviewer 2 for spotting this oversight and added this information to all SI Figure captions where it was missing.

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